PROJECT TITLE

Exploring The Relationship Between Economic Development and Health Outcomes: A Global Analysis Using the World Development Indicators (WDI) Dataset

Objective

To analyze the relationship between GDP per capita, GDP growth, health expenditure, and life expectancy across countries, and determine how economic growth and healthcare investment influence population health.

Problem statement

Economic growth is often associated with improvements in healthcare and living standards. However, the extent to which higher income translates into better health outcomes varies across countries.

This project aims to investigate how economic performance relates to life expectancy and healthcare expenditure while identifying countries and regions that perform exceptionally well or poorly.

Data overview

The World Development Indicators (WDI) dataset is a comprehensive collection of development statistics covering over 200 countries and territories. It includes indicators on economic performance, education, healthcare, trade, population, environment, and technology.

Indicators Used

- GDP per capita (current US$)
- GDP growth (annual %)
- Life expectancy at birth, total (years)
- Life expectancy at birth, female (years)
- Life expectancy at birth, male (years)
- Current health expenditure (% of GDP)

Data Cleaning and Preparation

- The following data preparation steps were performed in Excel Power Query to ensure the dataset was suitable for analysis:
- Imported the World Development Indicators (WDI) dataset into Excel.
- Promoted the first row to serve as the column headers.
- Filtered the dataset to retain only the indicators relevant to the study
- Unpivoted the year columns to transform the dataset from a wide format into a long format, resulting in Year and Indicator Value columns for easier analysis and visualization.
- Changed the Year column data type to Whole Number.
- Changed the Indicator Value column data type to Decimal Number.
- Checked for missing values and retained only records with valid data for the selected indicators.
- Removed duplicate records to improve data quality.
- Created a relationship between the WDI Data table and the WDI Country table using the Country Code field to enrich the dataset with regional information.
- Validated the data to ensure consistency in country names, years, and indicator values before building the dashboard.

Analysis Methodology

The analysis was conducted using Microsoft Power BI. The following analytical techniques were applied:

Descriptive Statistics: Used KPI cards to summarize key indicators such as average GDP per capita, average GDP growth, average life expectancy, and average health expenditure.

Trend Analysis: Used line charts to examine changes in GDP growth, life expectancy, and health expenditure over time.

Comparative Analysis: Compared countries and regions using bar and column charts to identify differences in economic and health indicators.

Regional Analysis: Analyzed indicators across World Bank regions to identify regional disparities.

Geographic Analysis: Used a filled map to visualize the geographical distribution of selected indicators.

Scatter Plot Analysis: Examined the relationship between GDP per capita and life expectancy, and between health expenditure and life expectancy.

KPI Analysis: Displayed key summary metrics to provide an overview of global economic and health performance.

Key findings

Health analysis:

- Global life expectancy has generally increased over time.
- Female life expectancy is consistently higher than male life expectancy across most countries.
- North America has the highest average life expectancy, while Sub-Saharan Africa has the lowest, indicating significant regional differences in health outcomes.

Economic analysis:

- GDP fluctuated between 2000 and 2024, highlighting periods of economic growth and slowdown
- The Top 10 countries recorded significantly higher GDP per capita than the global average
- North America has the highest average GDP, while Sub-Saharan Africa has the lowest.

Health expenditure:

- Health expenditure fluctuated between 2000 and 2024, showing changes in countries' investment in healthcare over time.

- The top 10 countries recorded the highest health expenditure as a percentage of GDP, indicating greater investment in healthcare.
- North America recorded the highest average health expenditure, while Sub-Saharan Africa recorded the lowest, reflecting regional differences in healthcare investment

<u>Relationship analysis:</u>

- Countries with higher GDP per capita generally recorded higher life expectancy, indicating a positive relationship between economic prosperity and health outcomes.
- Countries with higher health expenditure generally had higher life expectancy, suggesting that greater investment in healthcare is associated with better population health

Insights

- Life expectancy improved steadily between 2000 and 2024, indicating overall progress in global health outcomes, despite a temporary decline around 2021.
- Females consistently have a higher life expectancy than males across most countries, suggesting persistent differences in longevity by gender.
- North America recorded the highest average life expectancy and GDP per capita, while Sub-Saharan Africa recorded the lowest, highlighting significant regional disparities.
- Countries with higher GDP per capita generally have higher life expectancy, suggesting that economic prosperity is associated with better health outcomes.
- Higher health expenditure is generally linked to higher life expectancy, indicating that investment in healthcare contributes to improved population health.
- GDP growth fluctuated over the study period, reflecting changing global economic conditions and periods of economic expansion and slowdown.
- Large differences in GDP per capita and health expenditure across countries and regions suggest that economic resources and healthcare investment are unevenly distributed worldwide.

Recommendations

- Increase investment in healthcare in regions with lower life expectancy to improve access to quality health services
- Sustainable economic growth should be promoted through policies that improve productivity and income levels.
- Provide greater support to low-performing regions to reduce disparities in health and economic development
- Improve access to quality healthcare especially in regions with low life expectancy and slow economic growth

Conclusions

This project shows that regions with higher GDP per capita and greater health expenditure generally have higher life expectancy, indicating a positive relationship between economic development and health outcomes. The findings emphasize the importance of investing in health care and promoting sustainable economic growth to improve quality of life and reduce regional disparities

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